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Master Thesis Proposal

Controllable Text Generation Using Langevin Dynamics

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1 Introduction

Large Language Models (LLMs) have achieved unprecedented performance in text generation and reasoning. However, despite their fluency, standard autoregressive (AR) decoding provides limited mechanisms for fine-grained, inference-time control. For example, if an autoregressive model generates a fluent but logically inconsistent or toxic phrase, it possesses no mechanism to revise earlier tokens without discarding and regenerating the entire subsequent sequence. Furthermore, enforcing constraints on full sequences is intractable for autoregressive models, as it requires generating many full sequences in order to pick the one that fulfills the constraints best. While the probabilistic nature of AR decoding is well understood, steering the output to satisfy complex global constraints (such as sentiment, style, or logical consistency) without expensive retraining remains a significant challenge.

Controllable Text Generation (CTG) [Hu et al., 2018, Zhang et al., 2023] aims to ensure model outputs adhere to specific constraints while maintaining fluency. Alignment techniques like Reinforcement Learning from Human Feedback (RLHF) can be used to control autoregressive generation. However, they require a preference dataset and are a "static". More specifically, aligned models cannot easily adapt to new user-defined constraints at runtime without collecting further data that reflects the new preferences and without further fine-tuning. A promising alternative is Energy-Based Model (EBM) guidance, where decoding is treated as an optimization problem. Energy-based decoding methods generate sequences by iteratively refining the whole sequence, often using gradient information as guidance (Langevin Dynamics). Global constraints or guidance signals can be integrated naturally into this framework by adding terms that measure the quality of a sequence to the energy function.

Recently, methods inspired by Langevin Dynamics have been proposed to navigate the text generation landscape. However, many existing approaches adopt Langevin-inspired updates but deviate from classical Stochastic Gradient Langevin Dynamics (SGLD) assumptions [Welling and Teh, 2011] regarding noise scaling, step-size annealing, or Metropolis-Hastings correction. Consequently, these methods often function more like "noisy gradient descent" rather than true posterior sampling, potentially compromising the trade-off between control adherence and text diversity.

This thesis proposes to investigate energy-based decoding techniques that allow for the flexible incorporation of sequence-level controls. Specifically, we focus on implementing and evaluating gradient-guided sampling frameworks, such as Langevin Dynamics. By systematically comparing optimization in discrete token spaces versus continuous embedding spaces, we seek to determine if these methods can improve the reliability and semantic coherence of controllable text generation.

2 Literature Review

The field of Controllable Text Generation (CTG) has evolved from simple heuristic-based filtering to complex probabilistic modeling. This section categorizes existing approaches based on the stage of intervention and the nature of the applied constraints, followed by a detailed examination of Langevin Dynamics as the theoretical foundation for this research.

2.1 Taxonomy of Controllable Text Generation

CTG methods can be classified along two primary dimensions: the *stage of intervention* (when the control is applied) and the *granularity of the constraint* (what is being controlled).

2.1.1 Stage of Intervention

Training-based Alignment These methods alter the model parameters θ to inherently bias the model toward desired attributes.

- **Retraining and Supervised Fine-tuning (SFT):** Early approaches involved fine-tuning pre-trained models on attribute-specific datasets [Alhafni et al., 2024]. However, obtaining high-quality, domain-specific datasets is often difficult and expensive. Furthermore, SFT can lead to mode collapse, where the model generates high-confidence but repetitive text, sacrificing diversity. SFT alone often fails to guarantee strict adherence to complex sequence-level constraints.
- **Reinforcement Learning (RLHF):** Modern alignment techniques use Reinforcement Learning from Human Feedback to optimize a learned reward signal [Ouyang et al., 2022, Li et al., 2024]. While effective, these methods often suffer from the *alignment tax*, a reduction in output diversity, and are computationally expensive to adapt to new, dynamic constraints at runtime.

Inference-time Guidance (Decoding-time) These methods freeze the model parameters and intervene during the decoding process. This paradigm is the primary focus of this thesis.

- **Prompt Engineering:** Utilizing soft or hard prompts to steer generation [Ouyang et al., 2022].
- **Classifier Guidance:** Leveraging external classifiers to adjust token probabilities. Methods such as Plug and Play Language Models (PPLM) [Dathathri et al., 2020] and FUDGE [Yang and Klein, 2021] modify hidden states or predictive distributions based on attribute classifiers.
- **Contrastive Decoding:** Approaches including GeDi [Krause et al., 2021] and DExperts [Liu et al., 2021] use expert and anti-expert language models to contrastively steer probability mass away from undesirable attributes.

2.1.2 Granularity of Constraints

Constraints in CTG are generally categorized by the scope of their impact on the generated sequence.

Token-Level Constraints These are local constraints applied at specific decoding steps. Examples include hard lexical constraints (forcing inclusion or exclusion of specific words) or syntactic constraints. NeuroLogic Decoding [Lu et al., 2021] and K2T [Pascual et al., 2021] operate at this level, ensuring the presence of specific keywords in the output.

Global-Level Constraints These constraints apply to holistic properties such as sentiment, topic, or toxicity. Global constraints are difficult to enforce autoregressively because their evaluation often depends on the completed sequence, rendering greedy, next-token prediction insufficient. This thesis focuses primarily on global constraints, where energy-based optimization is most effective.

2.2 Theoretical Foundations: Langevin Dynamics

To address the limitations of autoregressive decoding in satisfying global constraints, recent work has turned to sampling methods rooted in statistical physics and Bayesian inference.

2.2.1 Stochastic Gradient Langevin Dynamics (SGLD)

It is important to note that Langevin Dynamics is inherently defined for continuous distributions. The foundational work of Welling and Teh [Welling and Teh, 2011] bridges optimization and sampling. In Bayesian learning, one seeks to sample from the posterior distribution

$$p(\theta \mid X) \propto p(X \mid \theta) p(\theta). \quad (1)$$

Stochastic Gradient Langevin Dynamics (SGLD) updates parameters according to

$$\Delta\theta_t = \frac{\epsilon_t}{2} \left(\nabla \log p(\theta_t) + \sum_{i=1}^N \nabla \log p(x_i \mid \theta_t) \right) + \eta_t, \quad (2)$$

where ϵ_t is the step size and $\eta_t \sim \mathcal{N}(0, \epsilon_t)$ is Gaussian noise.

In the context of controlled generation, the term $\sum_{i=1}^N \nabla \log p(x_i \mid \theta_t)$ represents the data likelihood (token probabilities). To adapt this framework for controllable generation, we augment the update rule by adding the gradient of an unnormalized, differentiable constraint function (such as a sentiment classifier), which actively guides the sampling toward the desired attribute.

Welling and Teh showed that by annealing ϵ_t toward zero, the injected noise enables convergence to the true posterior distribution rather than collapsing to a single Maximum A Posteriori (MAP) solution.

However, a major bottleneck in applying this to text is that discrete token spaces do not naturally possess gradients. While Langevin-based solutions work well in continuous spaces, applying them to text requires overcoming this non-differentiable nature, which is a key focus of this research.

2.3 Energy-Based Models for Text

Inspired by SGLD, Energy-Based Models (EBMs) frame text generation as sampling from a distribution defined by an energy function $E(x)$:

$$P(x) \propto \exp(-E(x)), \quad (3)$$

where x denotes a text sequence and lower energy corresponds to better satisfaction of constraints such as fluency or sentiment.

COLD Decoding [Qin et al., 2022] and MUCOLA [Kumar et al., 2022] apply this framework to large language models by defining the energy as a combination of the model likelihood and differentiable constraint functions. These methods perform iterative updates in the continuous embedding space of the language model.

Research Gap Despite their empirical success, many existing EBM-based methods trade theoretical guarantees for empirical stability. In particular, they often omit the Metropolis–Hastings correction step or fail to properly anneal the noise schedule, resulting in dynamics closer to noisy gradient descent than true sampling. Additionally, the discrete nature of text necessitates ad-hoc de-quantization strategies to reconcile continuous updates with token-level outputs.

This thesis builds upon the Discrete Langevin Sampler proposed by [Zhang et al., 2022], which formulates gradient-based proposals directly for discrete distributions. We investigate whether a more mathematically faithful application of Langevin Dynamics can yield improved control and stability in text generation.

2.4 Competitive Landscape: Diffusion Models

Parallel to Langevin-based EBMs, diffusion models have emerged as a non-autoregressive alternative for CTG. DiffuSeq [Gong et al., 2023] and EDLM [Xu et al., 2025] apply forward and reverse diffusion processes in embedding space. While diffusion and Langevin dynamics are closely related, diffusion models effectively learn the score function used in Langevin sampling; they typically require training separate neural networks to reverse the noise process.

In contrast, the Langevin-based approach explored in this thesis enables zero-shot transfer. It operates directly on the latent space of a frozen, pre-trained language model at inference time, requiring only the definition of an energy function. This offers a more lightweight and flexible mechanism for controllable generation without the need for task-specific parameter updates.

3 Goal and research questions

This thesis investigates contemporary sampling and generation techniques for controlled text generation. The research evaluates how effectively these methods adhere to predefined constraints while maintaining linguistic accuracy. Furthermore, the study explores the internal mechanics of the Llama 3 (8B) model, specifically focusing on its gradient signals and embedding distributions. Through structured experimentation utilizing benchmark datasets and metrics, this work addresses a critical gap identified in the current literature, leading to the following research questions:

RQ1 (Core Methodology): Training-Free Methodological Fidelity and Optimization Space Does enforcing training-free, theoretically correct Langevin noise scaling and annealing improve posterior coverage and stability compared to heuristic approximations?

- **RQ1a** How does the efficiency and text quality of a gradient-based proposal for discrete variables (using embedding gradients) compare to Langevin sampling performed in the continuous embedding space $\mathbf{S} \in \mathbb{R}^{L \times D}$?

- **RQ1b** To what extent does the Metropolis–Hastings (MH) correction step mitigate the discretization bias observed in methods such as COLD [Qin et al., 2022] or MU-COLA [Kumar et al., 2022]?

RQ2 (Application): Robustness and Quality–Control Trade-off Can the proposed rigorous Langevin frameworks satisfy predefined control conditions while maintaining high text quality?

- **RQ2a** How effectively can Langevin-based energy functions satisfy global constraints (e.g., sentiment) while maintaining the standards of fluency and diversity (measured via Perplexity and MAUVE) required for modern LLMs?
- **RQ2b** In the context of gradient-guided editing (text infilling), does energy-based sampling recover semantic coherence more accurately than standard autoregressive decoding?

RQ3: Improving Sampling Efficiency

- **RQ3a** How can we improve upon the efficiency of energy-based methods? Specifically, can we mitigate the computational cost of sampling by incorporating a lightweight training step (e.g., amortized inference or GFlowNet-inspired objectives) to better guide the proposal distribution?

4 Methods

The objective of this research is to move beyond "noisy gradient descent" and implement a mathematically sound sampling framework for Controllable Text Generation (CTG). The study will focus on Energy-Based Model (EBM) guidance, where the goal is to sample from a target distribution $p(x) \propto \exp(-E(x))$ defined by an energy function $E(x)$.

4.1 Baseline 1 - Discrete Langevin Sampler (DLS)

To address RQ2, this work will implement the gradient-based proposal for discrete variables proposed by [Zhang et al., 2022]. Unlike standard Langevin which requires a continuous space, this sampler operates in the discrete token space Θ .

Gradient Approximation Since the token space Θ is discrete, we cannot directly compute the gradient $\nabla U(\theta)$. Instead, we utilize gradients of the energy function with respect to the continuous token embeddings.

We define a local proposal distribution $q(\theta' | \theta)$ by scoring candidate tokens based on the inner product between their embeddings and the energy gradient, thereby biasing the selection toward tokens that minimize the energy.

Transition Kernel: The next discrete state θ' is sampled based on the proposal distribution:

$$q(\theta' | \theta) = \frac{\exp(-\frac{1}{2\alpha} \|\theta' - \theta - \frac{\alpha}{2} \nabla U(\theta)\|_2^2)}{Z_{\Theta}(\theta)}$$

Normalizing Constant: The term $Z_{\Theta}(\theta)$ is calculated by summing over the discrete vocabulary space Θ .

Conditioning: To enable controllable generation, the project will incorporate differentiable constraints C (e.g., sentiment scores) into the energy function

$$U(\theta) = -\log p(\theta) + \lambda C(\theta)$$

, where λ is a Lagrange multiplier for the constraint.

Here, Θ can be any space without affecting q being a valid proposal. As a special case, when $\Theta = \mathbb{R}^d$, it follows that $Z_{\mathbb{R}^d}(\theta) = (2\pi\alpha)^{d/2}$ and recovers the Gaussian proposal in the standard Langevin algorithm. When Θ is a discrete space, we naturally obtain a gradient-based proposal for discrete variables.

4.2 Baseline 2 - Continuous Langevin Sampler (CLS)

This method will address RQ1 and RQ2 by performing optimization in the embedding space of the Large Language Model (LLM). The sequence of embeddings $S \in \mathbb{R}^{L \times D}$ will be treated as continuous latent variables.

The methodology involves converting the embedding of the output sequence to a "soft sequence" and subsequently optimizing it using Langevin dynamics.

Core Idea. The corrupted segment embedding

$$S \in \mathbb{R}^{L \times D}$$

is treated as continuous latent variables, where L is the number of infilling positions and D is the embedding dimension.

Langevin Step (theoretical form).

$$s_{t+1} = s_t + \eta_t \nabla_s \log \pi(s_t) + \sqrt{\eta_t} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I).$$

Equivalent Interpretation. First compute a Gaussian proposal mean

$$m_t = s_t + \eta_t \nabla_s \log \pi(s_t).$$

Then sample

$$s_{t+1} \sim \mathcal{N}(m_t, \eta_t I).$$

4.3 Metropolis-Hastings (MH) Correction

To address RQ1, we integrate an MH correction step. Standard Langevin-inspired methods (like [Qin et al., 2022]) omit this, leading to discretization bias where the sampler may drift away from the true posterior. This involves calculating an acceptance probability:

$$A(\theta'|\theta) = \min \left(1, \frac{p(\theta')q(\theta|\theta')}{p(\theta)q(\theta'|\theta)} \right)$$

This step is intended to restore asymptotic correctness to the target distribution $p(\theta)$, ensuring that the sampler explores the true posterior rather than simply maximizing the objective.

4.4 Gibbs Sampling for Token Refinement

To provide a comparison for the joint update methods, Gibbs sampling will be implemented. This method updates one token at a time while keeping others fixed:

$$x_i \sim p(x_i|x_{-i}, \text{conditions})$$

The study will evaluate the trade-offs of this approach, specifically comparing its potentially slower convergence against the parallelizable nature of Langevin Dynamics.

4.5 Autoregressive (AR) Decoding Baseline

To answer RQ3, standard Autoregressive (AR) decoding methods, such as Top-P or Beam Search, will be utilized as a performance floor. This baseline represents the model’s natural generation without energy-based intervention, allowing for a measurement of the ”cost” of control regarding fluency and diversity.

5 Experiment

5.1 Text Infilling and Token Recovery

We will evaluate the models on a ”flipped token recovery” task using the CommonGen and ROCStories datasets. This serves as a diagnostic task: random tokens in a sequence will be masked or replaced, and the method must utilize the energy gradient from the surrounding context to recover semantic coherence.

5.2 Controlled Generation

We will generate text conditioned on Sentiment (Positive/Negative) and Toxicity (Non-toxic) using the IMDb and RealToxicityPrompts datasets.

5.3 Evaluation Metrics

The following list represents a comprehensive set of potential metrics. During the course of the thesis, we will select the most relevant subset to demonstrate the performance of our method.

- **Fluency:** Perplexity (PPL) computed using a larger, held-out language model (e.g., Llama-3-70B).
- **Distributional Quality:** MAUVE scores to measure how well the generated text distribution matches human-written text, ensuring that the sampler is not merely exploiting the reward model.
- **Control Adherence:** Classification accuracy of the generated text measured using an external classifier.
- **Diversity:** Self-BLEU and Distinct- n scores to capture lexical and structural diversity among generated samples.

- **LLM-based Evaluation:** Using a strong instruction-tuned model (e.g., GPT-4 or Llama-3-70B-Instruct) as a judge to rate the coherence and constraint satisfaction of the outputs.
- **Efficiency:** Number of Function Evaluations (NFE) to quantify the latency cost of Langevin sampling compared to autoregressive decoding.

6 Timeline

Month	Working Packages / Milestones
Month 1: Literature Review & Baseline Setup	• Literature review: Energy-based decoding, diffusion/LD for LLMs • Define evaluation metrics • Implement initial Langevin baseline
Month 2: Implementation & Baselines	• Efficient Langevin-like sampling implementation (large sequence support) • Implement baselines: Metropolis-Hastings, Gibbs sampling, AR decoding, Supervised alignment + AR decoding (if time permits).
Month 3: Token Recovery / Infilling Experiments	• Experiments on text infilling / flipped token recovery • Train auxiliary critique/guidance models (e.g., sentiment/mood classifier)
Month 4: Energy Decoding Framework	• Formulate unified energy-based decoding framework and key problems in closed energy based decoding framework • Controlled generation experiments (using trained guidance models)
Month 5: Optimization and Framework Extension	• Focusing on improving sampling efficiency and addressing key bottlenecks identified in Month 4
Month 6: Finalization & Writing	• Final experiments, documentation • Thesis writing and submission

Table 1: Proposed Thesis Timeline

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